

## Problem-Solution Fit

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| <b>Date</b>          | 02 July 2026                                                       |
| <b>Team ID</b>       | xxxxxxx                                                            |
| <b>Project Name</b>  | <b>OptiCrop: Smart Agricultural Production Optimization Engine</b> |
| <b>Maximum Marks</b> | 5 Marks                                                            |

### Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

|                                                                                                                                                                                                  |                                                                                                                                                                                                              |                                                                                                                                                                                                                             |
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| <b>1. CUSTOMER SEGMENT(S) [CS]</b><br><br>Farmers, agricultural researchers, agribusiness organizations, and policymakers                                                                        | <b>6. CUSTOMER LIMITATIONS [CL]</b><br><br>Limited access to modern agricultural tools, lack of technical knowledge, dependency on external advice                                                           | <b>5. AVAILABLE SOLUTIONS PROS &amp; CONS [AS]</b><br><br>Manual farming knowledge, traditional agricultural advisory systems<br><b>Pros:</b> Easy to access<br><b>Cons:</b> Not data driven, less accurate, time consuming |
| <b>2. PROBLEMS / PAINS + ITS FREQUENCY [PR]</b><br><br>Manual crop selection leads to low yield and poor decision making due to lack of datadriven guidance                                      | <b>9. PROBLEM ROOT/CAUSE [RC]</b><br><br>Lack of intelligent systems to analyze data for accurate crop prediction                                                                                            | <b>7. BEHAVIOR + ITS INTENSITY [BE]</b><br><br>Farmers frequently rely on seasonal experience and external suggestions for crop selection decisions                                                                         |
| <b>3. TRIGGERS TO ACT [TR]</b><br><br>Need for better crop yield, climate variability, increasing demand for smart farming solutions                                                             | <b>10. YOUR SOLUTION [SL]</b><br><br>Develop an ML-based Smart Crop Recommendation System using Python and Flask that analyzes soil nutrients and environmental conditions to predict the most suitable crop | <b>8. CHANNELS OF BEHAVIOR [CH]</b><br><br><b>ONLINE:</b> Agricultural websites, mobile farming apps, government portals<br><br><b>OFFLINE:</b> Local agricultural officers, farming consultants, cooperative societies     |
| <b>4. EMOTIONS BEFORE/AFTER [EM]</b><br><b>Before:</b> Confusion, uncertainty, dependency on guesswork.<br><b>After:</b> Confidence, clarity, improved decision-making, trust in recommendations |                                                                                                                                                                                                              |                                                                                                                                                                                                                             |